

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 64

THE QUINN CENTRIFUGAL TRENCH-GUTTER ZERO-G FLUID STABILIZATION

Spin the tank — the liquid climbs the walls, the bubbles flee center
the gutter sump keeps the pump's lock on pure liquid



liquid on the rim, bubbles in the middle, intake in the gutter

ROTATING TRACK RINGS — VERTICAL TRENCH GUTTER — $a = \omega^2 r$

A PUMP THAT SWALLOWS A BUBBLE IS A PUMP THAT DIES

targets, not absolutes — the bench quantifies, seated 14 August

GP-IM-064

Revision v1.0 — 22 August 2026

65 of 290

VETTED: ROTATING-FLUID PASS — QUANTIFY BY TEST

PRIMARY AUTHOR & INVENTOR: ARCHER QUINN — AEROSPACE DESIGN ENGINEER, NPHD

CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 64

PHASE 64: THE QUINN CENTRIFUGAL TRENCH-GUTTER ZERO-G FLUID STABILIZATION — STATUS: CURRENT. A pump that swallows a bubble is a pump that dies. Phase 64 mounts the Phase 22 Babushka internal fuel cells and all operational tanks onto low-power rotating track rings: sustained centrifugal containment pins the liquid mass to the perimeter walls, air pockets and bubbles consolidate at the rotational center away from the fluid lines, and the extraction intake draws from a vertical trench gutter, keeping an unbroken mechanical lock on pure liquid and preventing pump cavitation in zero-gravity. The vet graded the rotating-fluid principle PASS; the absolute claims — 100% containment, entirely bubble-free — were softened 14 August on his order ('replace as vet suggested') to design targets that the bench must quantify.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-064, v1.0 — 22 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the stabilization law as seated (contents line, the shared group objective, the design body — verbatim) — §2 why it works (centrifugal settling and the gutter — graded) — §3 the system in the tanks — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 64: **Centrifugal Trench-Gutter Zero-G Fluid Stabilization**. Solves the floating zero-gravity liquid chaos defect. Mounts primary storage tanks on a low-power axle-less turntable track, using low-RPM rotation to pin of the fluid volume smoothly against a concave peripheral trench gutter to keep lines bubble-free. This was the basis for multiple designs from the toilet to the coffee cups and drink bottles.'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Elimination of Solar Sandblasting & Microgravity Fuel-Slosh Pump Cavitation [the shared Phase 62-64 group objective, seated at the head of the group]'

* Rotational Kinetic Equilibrium: Inner fuel cells of the Phase 22 Babushka tanks and all operational tanks are mounted on low-power rotating track rings to override zero-G fluid floating. * Centrifugal Containment: Slow-spin rotation pins the liquid fuel mass smoothly against the outermost perimeter walls of the tank, forcing all air pockets to consolidate into a harmless dead-zone in the center. * Vertical Trench: Fluid is captured within a continuous vertical trench gutter machined along the interior outer circumference. The gutter narrows to feed the propulsion pumps, ensuring an unbroken lock on pure liquid to eliminate pump cavitation during high-torque structural pivots. The full-pin and zero-cavitation claims stand as design targets to be quantified by test — gas-liquid mixtures, partial filling, acceleration transients, sloshing and wetting behaviour all matter.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE CENTRIFUGAL SETTLING, AFFIRMED: $a = \omega^2 r$ — rotation pins the denser phase (liquid) to the rim and drives the lighter phase (gas) to the center; this is how every centrifuge and every spin-settled propellant depot works. The principle graded PASS without argument.

THE CAVITATION TARGET, AFFIRMED AS TARGET: unbroken liquid at the intake is the correct engineering goal — cavitation destroys impellers. The vet's flag was on the absolutes: '100%' and 'bubble-free' are claims no rotating tank fully delivers at every flow rate; seated 14 August, the claims now stand as design targets to be quantified by test, not inventor error.

THE TRENCH GUTTER, AFFIRMED AS GEOMETRY: a vertical gutter at the rim gives the intake a sump — the same reason bilge pumps sit in a well. The gutter concentrates what the rotation pins; the pump never sees the center.

THE LOW-POWER POSTURE, AFFIRMED AS MASS DOCTRINE: track-ring rotation at low power buys pump reliability — the energy cost is owed to the budget, the mechanical payoff is immediate.

§3 — THE SYSTEM IN THE TANKS

- THE RINGS: Phase 22 Babushka internal fuel cells and all operational tanks on low-power rotating track rings.
- THE SETTLING: sustained centrifugal containment pins liquid to the perimeter walls; air pockets and bubbles consolidate at the rotational center.
- THE GUTTER: the vertical trench gutter at the rim carries the extraction intake — an unbroken mechanical lock on pure liquid.

- THE TARGETS: containment fraction and bubble-free operation stand as design targets, quantified by the owed bench (14 August).

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE SUMP DOCTRINE (seated with the gutter): give the intake a well and the pump never drinks air — the oldest plumbing answer, spun.
- THE TARGETS-NOT-ABSOLUTES DOCTRINE (the vet's repair, seated 14 August): the design aims at the absolute; the chapter claims only what the bench has measured.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Mount the rings: a Phase 22 cell on a rotating track rig; drive power and bearing life logged.
- STEP 2 — Bench the settling: liquid fraction at the rim versus rotation rate and fill level; bubble migration to center imaged and published.
- STEP 3 — Fit the gutter: the vertical trench gutter machined into the cell; intake draw tested across rotation and flow rates — this is the quantification the vet asked for.
- STEP 4 — Break the pump on purpose: throttle, slosh and stall transients run until cavitation appears; the envelope mapped and the operating floor written above it.
- STEP 5 — Budget the power: ring-drive load against the ship's electrical baseline (Phase 65 territory).
- STEP 6 — Publish the numbers: containment fraction, cavitation envelope, drive budget — the targets as measured.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 22 — the Babushka scissor-brace tanks (GP-IM-022): the cells the rings rotate.
- PHASE 61 — the centrifugal sluice (GP-IM-061): the same separation family in the fuel lines.
- PHASES 62/63 — the filters and the acoustic protocols (GP-IM-062/063): the shared 62-64 group.
- PHASE 65 — the inverter telemetry grid (GP-IM-065): the electrical baseline the ring drive budgets against.

§8 — DO-NOT-BUILD

- NEVER claim 100% — containment is a measured fraction at an operating point; absolutes died 14 August.
- NO static intakes in a spinning tank — the intake lives in the gutter or it drinks bubbles.
- NEVER run the pump below the measured floor — the cavitation envelope is law once benched.
- NO free-spinning assumptions — the ring drive is a budget line; budget it.

§9 — OWED

THE BENCH, OWED (the vet's item): containment fraction and bubble migration versus rotation rate and fill level; the cavitation envelope mapped; the ring-drive power budget.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-064, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). Built 22 August 2026, seventh of the new 'next ten' shift. First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557 and 561): 'ok next 10' / 'ok next ten'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 64